1. What is our project?

This is a **web-based customer support system** where:

- Users can **ask questions**
- Users can **raise complaints/tickets**
- A **chatbot automatically answers basic queries**
- **AI analyzes the user's message** (happy/angry/neutral)
- Admin can see all tickets and solve them

In short:

It is like a small version of Amazon/Flipkart customer support.

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2. Why are we making this project?

Most companies receive:

- Too many support queries
- Repeated questions
- No idea which customer is upset
- No system to prioritize urgent cases

Our project solves these issues by adding **AI + automation**.

It reduces manual work and improves customer satisfaction.

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3. How does the system start working?

1. User visits website
2. User logs in
3. User can either:
    - Chat with the chatbot, OR
    - Raise a support ticket
4. Chatbot answers simple questions
5. For more serious issues → Ticket is created
6. Admin logs in and solves the ticket
7. AI helps admin by showing sentiment and urgency

This is exactly how customer support works in real companies.

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4. What happens inside the system? (Internal Working – Simple)

The project has **three main parts** working together:

1. Frontend (Website Users See)

Built in React JS

- Login page
- Chatbot page
- Ticket form
- Admin dashboard

This is the **interface**.

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2. Backend (.NET API – The Brain of the System)

This part:

- Checks login
- Stores users
- Stores tickets
- Takes data from AI
- Sends results back to frontend

This is the **decision-maker**.

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4. AI Chatbot Service (Node.js – The Helper Part)

This part:

- Reads user message
- Gives basic reply
- Checks sentiment (angry/happy)
- Classifies ticket (technical/billing/general)

This is the **smart assistant**.

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4. Databases (Memory of the System)

- SQL Server → Stores **users & authentication**
- MongoDB → Stores **tickets & chats**

This is the **memory**.

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5. How does it work in real life? (Example)

Example scenario:

1. A customer types:
   "My order is delayed, I am frustrated."
2. Chatbot replies politely:
   "I understand your concern. Please wait while we create a ticket for you."
3. AI reads message:
   - Sentiment: **Negative**
   - Category: **Delivery Issue**
   - Priority: **High**
4. Ticket reaches admin with labels:
   - "High Priority"
   - "User upset"
5. Admin resolves issue faster.

=>This increases customer satisfaction.

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6. Why is this project useful for companies?

Because it:

- Saves time
- Reduces workload
- Improves customer experience
- Gives analytics to management
- Uses AI to understand customers

Companies like:

- Amazon
- Swiggy
- Jio
- Banks
  all use similar systems.

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7. Why this project is perfect for us?

Because it includes:

- Chatbot (NLP)
- Sentiment analysis (AI)
- Classification model
- Microservices architecture
- Real-time communication

This shows both:

- **Software engineering skills**
- **AI skills**